

Solution Requirements

Date	2 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system allows users to access the crop prediction application. (No login is required in the current version.)	Medium
2	Authorization levels	All users have the same access to use the crop prediction feature. No separate user roles are implemented.	Low

3	External interfaces	The system interacts with a trained Machine Learning model (model.pkl) and reads input data from the web interface.	High
4	Transactions processing	The system receives soil and weather values, processes them through the ML model, and returns the recommended crop.	High
5	Reporting	The system displays the recommended crop instantly on the result page.	Medium
6	Business rules, etc.	Crop recommendation is generated based on Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, pH, and Rainfall values.	High
7	Compliance to laws or regulations	The application does not collect or store personal user data and follows basic software development practicesThe application validates user input before generating the crop recommendation.	Low

8	<i>Other</i>	The application validates user input before generating the crop recommendation.	Medium
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Step 2: Non-Functional Requirements (NFR)

Performance & Speed	The system should generate crop recommendations quickly.	Response time < 2 seconds
Scalability	The system should support more users and additional crop datasets in future versions.	Supports 100+ simultaneous users
Security & Data Privacy	User input should be processed securely without storing personal information.	Input validation and secure request handling
Reliability & Availability	The application should operate reliably during normal usage.	99% availability during testing
Usability & Accessibility	The interface should be simple, responsive, and easy for farmers to use.	Easy navigation with clear labels and buttons
Others	The application should be easy to maintain, update, and deploy on different operating systems.	Compatible with Windows, Linux, and cloud platforms